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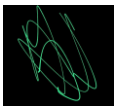
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COLLEGE OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
CSC 305 – Software Engineering | Semester 1 – 2026/2027 | Section 5MA1 | Group 3

Project Proposal

InternSync – Cooperative Training Coordination and Tracking System

1. Project Title and Team Information

Project title: InternSync – Cooperative Training Coordination and Tracking System

Team Member	Student ID	Signature	Date
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2. Project Background

Cooperative training (Co-op) and summer internship programs are a key step in preparing students for the job market: students spend a period inside a company and apply what they learned at the college. Each training cycle involves four parties: the **student**; the **company**, which hosts the student and assigns a training supervisor; an **academic supervisor** from the college who follows the student's progress and approves the grade; and the **co-op unit coordinator**, who manages vacancies, deadlines, and the official records.

Today these activities are coordinated through email, spreadsheets, and paper or PDF forms. Vacancies are circulated informally, weekly reports arrive as email attachments, company supervisors fill evaluation forms (usually only at the end of the period), and final grade sheets are compiled by hand. This works for a handful of students, but it does not scale and leaves no single, reliable record of what happened during a training cycle.

InternSync is proposed as one web platform that supports the whole training life cycle, from publishing vacancies and matching students to them, through weekly follow-up, evaluation, and grade approval, to the final grade sheets and a verifiable completion certificate.

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3. Problem Statement

The current process is not only inconvenient; each weakness below directly affects students, supervisors, or the accuracy of the final grades.

#	Problem	How it works today	Why it is a problem
P1	Blind, manual matching	Vacancies are circulated informally and coordinators match applicants to companies by hand.	Students apply without knowing whether their skills fit, which leads to mismatched placements and wasted applications. Coordinators spend time on manual work, and nobody can explain why a student fits a vacancy or what the student is missing.
P2	Untracked weekly reporting	Reports arrive as email attachments.	There is no reliable record of who submitted what and when. Late or missing reports are noticed only after the fact, and disputes cannot be settled without a timestamped history.
P3	Fragmented, late evaluation	Company supervisors fill paper or PDF forms, usually at the end of the training period.	There is no continuous performance record, so students get no feedback while they can still improve. Each company uses different criteria, which makes grades hard to compare fairly.
P4	No academic-integrity check	Academic supervisors read reports one by one.	Recycled or copied reports across students and across terms are hard to notice, which weakens the credibility of the grades.
P5	Slow, error-prone grading	Final grade sheets are compiled manually from scattered sources.	Results are delayed, transcription mistakes are possible, and the same manual effort is repeated every term.
P6	No outcome tracking or easy verification	There is no built-in way to record outcomes or to verify completion.	The college lacks data on how many placements turn into job offers, and third parties cannot easily verify that a student completed the training.

Opportunity: consolidate these scattered activities into one auditable system that enforces deadlines automatically, keeps a complete history of every submission, evaluation, and approval, and generates the official outputs directly from that history.

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4. Project Objectives

Each objective has a measure of success that can be checked during testing (by the end of the testing phase, Dec 2026). Priorities follow the Must / Should / Could split used in the scope (Section 7).

ID	Objective	Measure of success	Priority
O1	Deliver a role-based web platform (responsive PWA) for four user roles with secure authentication and permissions.	All Must-have functions (F1–F10) are implemented, and access-control tests show every role can reach only its permitted actions (0 failing permission tests).	Must
O2	Rank vacancies against each student's profile and explain every result.	On a labelled test set of at least 20 student profiles, at least one relevant vacancy appears in the top 5 for 80% or more of the profiles; each result lists matched and missing skills.	Must / Should
O3	Track weekly submissions and deadlines automatically.	100% of weekly deadlines are tracked; in testing, reminders are sent before the deadline and late submissions are flagged within 1 hour after it.	Must / Should
O4	Standardize company evaluations.	100% of periodic and final evaluations use one rubric (same criteria and an anchored 1–5 scale) for all companies.	Must
O5	Give academic supervisors a similarity indicator for reports.	On 10 seeded reports with 40% or more copied text, at least 9 are flagged and the matching passages are shown. The final judgment stays with the supervisor.	Should
O6	Enforce multi-stage approval and remove manual grade re-entry.	No grade becomes final until every required role has approved (verified by workflow tests); grade sheets for a 100-student test cohort are exported to PDF and Excel in under 60 seconds.	Must / Should
O7	Support Arabic and English.	100% of screens are available in both languages with correct RTL layout; matching works on both Arabic and English CVs.	Must
O8	Check usability with real users.	User acceptance testing with at least 8 participants covering all roles; at least 80% of test scenarios are completed without help.	Must
O9	Issue verifiable completion certificates with a QR code.	The public verification page confirms valid certificates and rejects tampered or revoked ones in all test cases.	Could
O10	Report placement outcomes.	The administration dashboard shows the placement-to-job-offer conversion rate per cycle and per company, based on end-of-term survey data.	Could

5. Proposed Solution

InternSync is a role-based web platform, delivered as a responsive Progressive Web App (PWA), that manages the complete co-op life cycle in one place. It is installable on a phone, works in Arabic and English, and uses AI only to **support** decisions; people always make the final decision.

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5.1 How the system works

1. The coordinator opens a training cycle and publishes vacancies (companies can also add vacancies).
2. The student builds a profile and uploads a CV. The system ranks vacancies by fit, explains each match, shows missing skills, and suggests a development plan.
3. The student applies and the coordinator confirms the placement.
4. During training the student submits weekly tasks and reports; the company supervisor confirms attendance weekly. Reminders and late flags are sent automatically.
5. The academic supervisor reviews reports with a similarity indicator and can request revisions.
6. The company supervisor submits periodic and final evaluations using one standard rubric.
7. Reports and grades pass through a multi-stage approval workflow; a grade becomes final only after every responsible role approves.
8. The system exports official grade sheets (PDF and Excel) and can issue a completion certificate with a QR code that anyone can verify.

5.2 AI components

- **Matching and explanation:** text embeddings of the CV and the vacancy description rank vacancies, with a rule-based fallback (skill keywords, specialization, GPA) so the feature still works without embedding data. Each result explains why it matches.
- **Skill-gap analysis and development plan:** highlights missing qualifications for a chosen vacancy and suggests learning steps.
- **Similarity detection:** compares submitted reports stored in the system and flags overlap for the supervisor's review.
- **Multilingual support:** a multilingual embedding model so Arabic and English CVs and reports are handled.

5.3 Tools and technologies

Layer	Technology
Frontend	React as a responsive PWA (installable, web push notifications, cached app shell); Arabic/English interface with RTL support.
Backend	Python with FastAPI: one REST API that also hosts the AI services (matching, similarity).
Database	PostgreSQL for relational data, with pgvector to store embeddings.
AI	Open-source multilingual text-embedding model running on the server, so CV data does not leave the system; rule-based fallback.
Security	JWT-based authentication and role-based access control.
Background jobs	A scheduler for reminders, late flags, and escalation.
Files and exports	File storage for CVs and report attachments; PDF and Excel generation with Arabic (RTL) support.

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6. Main Stakeholders

Stakeholder	Type	Needs and interests	Interaction with the system
Student	Primary user	Find suitable vacancies, know what skills are missing, submit reports easily, follow own progress and feedback.	Profile and CV, apply, weekly reports, view evaluations, receive certificate.
Company training supervisor	Primary user (external)	A simple way to confirm attendance and evaluate students; a reliable channel to reach qualified students.	Attendance confirmation, periodic and final evaluations, vacancies.
Academic supervisor (faculty)	Primary user	Review reports efficiently, notice copied work, approve grades with a clear record.	Report review, similarity indicator, revision requests, grade approval.
Co-op coordinator (college co-op unit)	Primary user	Run the co-op program for the college: manage cycles, vacancies, deadlines, and official outputs without manual work.	Cycle and vacancy management, placement confirmation, escalations, user accounts and roles (system administration in the first release), exports, dashboards.
College administration	Indirect	Accurate program statistics and placement outcomes.	Dashboards and summary reports.
Partner companies and employers	Indirect	Reach qualified students; verify that a student completed the training.	Vacancies; public certificate verification.

7. Project Scope

Functions are prioritized: **Must** = committed baseline, **Should** = planned for the project, **Could** = only if the schedule allows.

7.1 In scope

ID	Function	Priority
F1	Role-based accounts, secure authentication, and authorization for four roles: student, company supervisor, academic supervisor, and coordinator/administrator.	Must
F2	Student profile and CV upload with skill extraction; student and course data imported from CSV files.	Must
F3	Vacancy catalogue: creation and publication (company or coordinator), browsing, filtering, application, and placement confirmation by the coordinator.	Must
F4	Matching engine that ranks vacancies against the student profile (text embeddings, with a rule-based fallback).	Must
F5	Weekly task and report submission with file attachments, deadlines, and revision requests.	Must
F6	Weekly attendance confirmation by the company supervisor.	Must
F7	Periodic and final evaluation forms using one standard rubric for all companies.	Must
F8	Multi-stage approval workflow for report acceptance and grade approval.	Must
F9	Role-specific dashboards showing pending actions and status.	Must
F10	Arabic/English interface with RTL and LTR support.	Must

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ID	Function	Priority
F11	Match explanation, skill-gap analysis, and development plan.	Should
F12	Similarity detection between reports stored in the system, producing a flag for supervisor review.	Should
F13	Scheduled notifications (in-app and email): reminders, late flags, and escalation to the coordinator.	Should
F14	Export of grade sheets and term summary reports as PDF and Excel.	Should
F15	Progressive Web App features: installable app, web push notifications, cached app shell.	Should
F16	Digital completion certificate with a QR code and a public verification page (including revocation).	Could
F17	Mutual evaluation: the student evaluates the training entity.	Could
F18	End-of-term survey and placement-to-job-offer statistics.	Could
F19	Evaluation-leniency alert for companies with at least 5 evaluations.	Could

7.2 Out of scope

Excluded item	Reason
Live integration with the university student information system	Student and course data are imported through CSV files instead.
Native mobile application	The responsive PWA covers phone use.
Contracts, payroll, insurance, and legal or HR processing with companies	Outside the purpose of training coordination.
Plagiarism detection against the public internet or external databases	Similarity is computed only between reports stored in the system; the supervisor decides.
Detection of AI-generated text	Such detectors are unreliable and could accuse students unfairly.
Biometric or GPS-based attendance verification	Privacy concerns and extra complexity; attendance is confirmed by the supervisor.
Real-time chat between users	Communication is limited to notifications and comments on submissions.
Statistical calibration of grades across companies	Not enough data per company; a standard rubric is used instead.
Writing grades directly into the university system	Grades are exported as files only.

7.3 Assumptions and constraints

- The system is developed and demonstrated with sample or synthetic data for one training cycle.
- Users have internet access and a modern browser.
- AI results (matching, skill gaps, similarity) are advisory; people make the final decisions.
- The project must be completed within the course schedule, so scope is the flexible variable and time is fixed.

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8. Proposed Software Development Process

Approach: plan-driven development with incremental delivery. All major activities (requirements, design, implementation, testing, and documentation) are planned in advance with dated phases and defined deliverables, and progress is measured against this plan. This suits the project because the course fixes both the deadlines and the documents to be submitted. Implementation is delivered in three planned increments ordered by priority (Must, then Should, then Could), so a working, tested system exists early and, if the schedule slips, lower-priority functions are dropped instead of moving the deadlines.

An agile process such as Scrum relies on changing requirements and flexible time and scope, which does not match a fixed course schedule with graded deliverables. A strictly sequential Waterfall process would leave testing until the end; planned increments reduce that risk while keeping the plan-driven structure.

The detailed schedule, milestones, roles, resources, risks, and monitoring activities are given in the Software Project Management Plan (SPMP).

9. Initial Feasibility Considerations

Aspect	Assessment
Technical	All selected technologies are open source and well documented. One Python/FastAPI codebase hosts both the API and the AI services, PostgreSQL with pgvector stores embeddings next to the relational data, and the embedding model runs on the server, so CV data does not leave the system. A rule-based fallback keeps matching working if the model is unavailable. Tools that are new to some members will be evaluated during the design phase.
Economic	No licence costs: the tools are open source or free for students (GitHub, Figma). Development runs locally with Docker, and a free or low-cost cloud service is enough for the demo. The main cost is team effort, assumed at about 8 hours per member per week over 10 weeks (6 × 8 × 10 ≈ 480 person-hours).
Schedule	The timeline is tight, but feasible because the scope is split into Must, Should, and Could. If the project slips, Could functions are dropped first, then Should functions. The detailed schedule is in the SPMP.
Operational	Each role gets a simple dashboard that shows only its pending actions. CSV import avoids dependence on the university systems. Email is kept as a fallback for notifications, since web push on some phones requires the app to be installed. Acceptance testing with users from every role checks that the system fits the real process.
Legal, ethical, and privacy	CVs and reports contain personal data, so access is limited to authorised roles, a consent notice is shown, and personal-data protection requirements in Saudi Arabia (such as the PDPL) are considered in the design. AI outputs are advisory: the similarity indicator is not a verdict, and matching does not decide placement.

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